Classical Ciphers

Playfair Cipher

- Not even the large number of keys in a monoalphabetic cipher provides security
- One approach to improving security was to encrypt multiple letters
- The **Playfair Cipher** is an example invented by Charles Wheatstone in 1854, but named after his friend Baron Playfair.
- When it was first put to the British Foreign Office as a cipher, it was rejected due to its perceived complexity.

Playfair Cipher

- Later adopted as a military cipher due to it being reasonably fast to use, and it requires no special equipment.
- Providing a stronger cipher than a Monoalphabetic Substitution Cipher.
- Used in the Second Boer War, and both World War I and World War II to different degrees.
- No longer used by military forces due to the advent of powerful computers.
- In its time, it was a relatively secure cipher that was easy to implement quickly.

Encyption

- To encrypt using the Playfair Cipher, start by creating a 5x5 matrix.
- Fill the matrix with letters from a keyword first, followed by the remaining letters of the alphabet.
- The English alphabet has 26 letters, but the matrix only has 25 squares.
- To fit all letters, you typically combine "i" and "j" into one square or leave out "q.“

Encyption

- eg. using the keyword MONARCHY

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Encyption

We must now split the plaintext up into digraphs (that is pairs of letters). On each digraph we peform the following encryption steps:

1. If the digraph consists of the same letter twice (or there is only one letter left by itself at the end of the plaintext) then insert the letter "X" between the same letters (or at the end), and then continue with the rest of the steps.
2. If the two letters appear on the same row in the square, then replace each letter by the letter immediately to the right of it in the square (cycling round to the left hand side if necessary).

Encyption

3. If the two letters appear in the same column in the square, then replace each letter by the letter immediately below it in the square (cycling round to the top of the square if necessary).
4. Otherwise, each plaintext letter in a pair is replaced by the letter that lies in its own row and the column occupied by the other plaintext letter.

Example

Step 1: Choose a Keyword

Let's use the keyword **"MONARCHY"**.

Step 2: Create the 5x5 Matrix

We create the matrix by placing the letters of the keyword first, removing duplicates, and then filling in the remaining letters of the alphabet. We'll combine "I" and "J" in one cell.

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Example

Prepare the Plaintext

Let's encrypt the plaintext **"HELLO"**.

Break it into digrams (pairs of two letters). If a pair has two identical letters or is incomplete, add an "X" to separate them: **HE LX LO**.

Apply the Playfair Cipher Rules

According to Rule 4

- **HE** becomes **CF**.
- **LX** becomes **SU**.
- **LO** becomes **PM**.

Final Encrypted Text: "CFSUPM."

Example for your Practice

As an example we shall encrypt the

plaintext **"hide the gold in the tree stump"** keyphrase ***playfair example***.

We do this by setting out a 5x5 grid, and filling it with the alphabet, starting with the letters of the keyphrase, and ignoring any letters we already have in the square. We are also going to combine "I" and "J" in the square.

Solution

P	L	A	Y	F
I	R	E	X	M
B	C	D	G	H
K	N	O	Q	S
T	U	V	W	Z

We must now split the plaintext into digraphs. At this point it is a good idea to apply Rule 1, and split up any double letter digraphs by inserting an "x" between them.

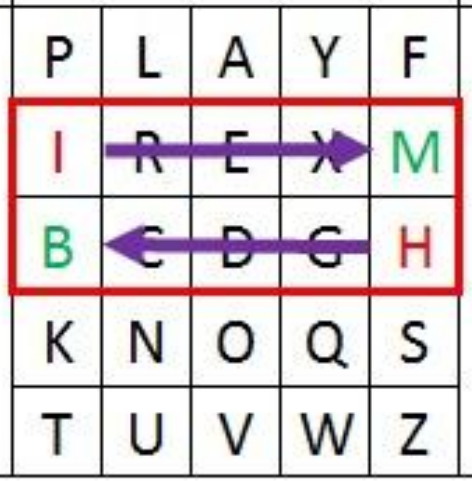
hi	de	th	eg	ol	di	nt	he	tr	ee	st	um	p

The intial split into digraphs.

hi	de	th	eg	ol	di	nt	he	tr	ex	es	tu	mp

The digraph split once we apply Rule 1, and remove any digraphs made from two of the same letter.

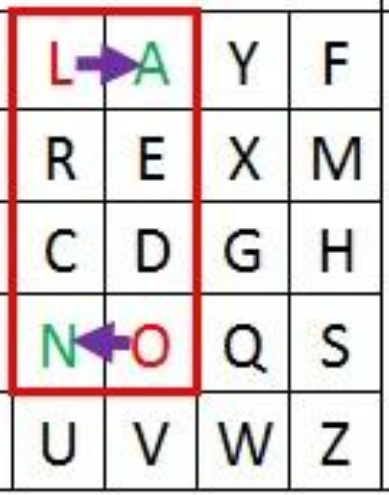
We now take each digraph in turn and apply rule 2, 3 or 4 as necessary. Each step is show below with a visual representation of what is done for each digraph.

Plaintext Digraph	Square	Rule	Ciphertext Digraph
hi		Rule 4: Rectangle	BM

Plaintext Digraph	Square					Rule	Ciphertext Digraph
de	P	L	A	Y	F	Rule 3: Same Column	OD
	I	R	E	X	M		
	B	C	D	G	H		
	K	N	O	Q	S		
	T	U	V	W	Z		

Plaintext Digraph	Square	Rule	Ciphertext Digraph
th		Rule 4: Rectangle	ZB

Plaintext Digraph	Square					Rule	Ciphertext Digraph
eg	P	L	A	Y	F	Rule 4: Rectangle	XD
	I	R	E	X	M		
	B	C	D	G	H		
	K	N	O	Q	S		
	T	U	V	W	Z		

Plaintext Digraph	Square	Rule	Ciphertext Digraph
ol		Rule 4: Rectangle	NA

Plaintext Digraph	Square					Rule	Ciphertext Digraph
di	P	L	A	Y	F	Rule 4: Rectangle	BE
	I	R	E	X	M		
	B	C	D	G	H		
	K	N	O	Q	S		
	T	U	V	W	Z		

Plaintext Digraph	Square					Rule	Ciphertext Digraph
nt	P	L	A	Y	F	Rule 4: Rectangle	KU
	I	R	E	X	M		
	B	C	D	G	H		
	K	N	O	Q	S		
	T	U	V	W	Z		

Plaintext Digraph	Square					Rule	Ciphertext Digraph
he	P	L	A	Y	F	Rule 4: Rectangle	DM
	I	R	E	X	M		
	B	C	D	G	H		
	K	N	O	Q	S		
	T	U	V	W	Z		

Plaintext Digraph	Square					Rule	Ciphertext Digraph
tr	P	L	A	Y	F	Rule 4: Rectangle	UI
	I	R	E	X	M		
	B	C	D	G	H		
	K	N	O	Q	S		
	T	U	V	W	Z		

Plaintext Digraph	Square					Rule	Ciphertext Digraph
ex	P	L	A	Y	F	Rule 2: Same Row	XM
	I	R	E	X	M		
	B	C	D	G	H		
	K	N	O	Q	S		
	T	U	V	W	Z		

Plaintext Digraph	Square					Rule	Ciphertext Digraph
es	P	L	A	Y	F	Rule 4: Rectangle	MO
	I	R	E	X	M		
	B	C	D	G	H		
	K	N	O	Q	S		
	T	U	V	W	Z		

Plaintext Digraph	Square					Rule	Ciphertext Digraph
tu	P	L	A	Y	F	Rule 2: Same Row	UV
	I	R	E	X	M		
	B	C	D	G	H		
	K	N	O	Q	S		
	T	U	V	W	Z		

Plaintext Digraph	Square	Rule	Ciphertext Digraph
mp	P L A Y F I R E X M B C D G H K N O Q S T U V W Z	Rule 4: Rectangle	IF

We can now take each of the ciphertext digraphs that we produced and put them all together.

BM	OD	ZB	XD	NA	BE	KU	DM	UI	XM	MO	UV	IF

We can now write out the ciphertext as a long string "BMODZBXDNABEKUDMUIXMMOUVIF" or split it into block of 5 "BMODZ BXDNA BEKUD MUIXM MOUVI F" or even give it the same layout as the original "BMOD ZBX DNAB EK UDM UIXMM OUVIF"

Decryption

- Decryption is nearly identical to the encryption process, except for rules 2 and 3 we must take the letters to the left and above respectively.
- Also, we remove any extra "X" in the decrypted text to reveal the final plaintext.

Decryption Example

We shall decipher the

ciphertext **"UA ARBED EXAPO PR QNX AXANR"**

which has been encrypted using the

Keyword: **example**

Firstly we must generate the Polybius Square which we are using, as shown to the right.

E	X	A	M	P
L	B	C	D	F
G	H	I	K	N
O	Q	R	S	T
U	V	W	Y	Z

The next step is to split the ciphertext into digraphs. There is no need to add any "X" in the decryption process as these will be revealed as we decrypt.

UA	AR	BE	DE	XA	PO	PR	QN	XA	XA	NR

Now we apply the rules as needed to each digraph in the ciphertext.

Ciphertext Digraph	Square					Rule	Plaintext Digraph
UA	E	K	A	M	P	Rule 4: Rectangle	we
	L	B	C	D	F		
	G	H	I	K	N		
	O	Q	R	S	T		
	U	V	W	Y	Z		

Ciphertext Digraph	Square					Rule	Plaintext Digraph
AR	E	X	A	M	P	Rule 3: Same Column	wi
	L	B	C	D	F		
	G	H	I	K	N		
	O	Q	R	S	T		
	U	V	W	Y	Z		

Ciphertext Digraph	Square					Rule	Plaintext Digraph
BE	E→	→X	A	M	P	Rule 4: Rectangle	lx
	L←	←B	C	D	F		
	G	H	I	K	N		
	O	Q	R	S	T		
	U	V	W	Y	Z		

Ciphertext Digraph	Square					Rule	Plaintext Digraph
DE	E	X	A	M	P	Rule 4: Rectangle	lm
	L	B	C	D	F		
	G	H	I	K	N		
	O	Q	R	S	T		
	U	V	W	Y	Z		

Ciphertext Digraph	Square					Rule	Plaintext Digraph
XA	E	X	A	M	P	Rule 2: Same Row	ex
	L	B	C	D	F		
	G	H	I	K	N		
	O	Q	R	S	T		
	U	V	W	Y	Z		

Ciphertext Digraph	Square	Rule	Plaintext Digraph
PO	E K A M P L B C D F G H I K N O Q R J T U V W Y Z	Rule 4: Rectangle	et

Ciphertext Digraph	Square					Rule	Plaintext Digraph
PR	E	X	A	M	P	Rule 4: Rectangle	at
	L	B	C	D	F		
	G	H	I	K	N		
	O	Q	R	J	T		
	U	V	W	Y	Z		

Ciphertext Digraph	Square					Rule	Plaintext Digraph
QN	E	X	A	M	P	Rule 4: Rectangle	th
	L	B	C	D	F		
	G	H	I	K	N		
	O	Q	R	S	T		
	U	V	W	Y	Z		

Ciphertext Digraph	Square					Rule	Plaintext Digraph
XA	E	X	A	M	P	Rule 2: Same Row	ex
	L	B	C	D	F		
	G	H	I	K	N		
	O	Q	R	S	T		
	U	V	W	Y	Z		

Ciphertext Digraph	Square					Rule	Plaintext Digraph
XA	E	X	A	M	P	Rule 2: Same Row	ex
	L	B	C	D	F		
	G	H	I	K	N		
	O	Q	R	S	T		
	U	V	W	Y	Z		

Ciphertext Digraph	Square					Rule	Plaintext Digraph
NR	E	X	A	M	P	Rule 4: Rectangle	it
	L	B	C	D	F		
	G	H	I	K	N		
	O	Q	R	S	T		
	U	V	W	Y	Z		

We now combine all the digraphs together.

we	wi	lx	lm	ex	et	at	th	ex	ex	it

So we get the message "we wilxl mexet at thex exit". When we remove the unnecessary "x"s we get a final plaintext of "we will meet at the exit". Note that we cannot just remove all the "x"s as one is part of the word "exit".

Small Quiz

Encrypt using PlayFair

- Original Text: Meet me at noon

Decrypt using PlayFair

- Cipher Text: CB NV NY SC

Security of Playfair Cipher

In the decryption example provided:

- Three identical digraphs "XA" are present in the ciphertext.
- All three "XA" digraphs decrypt to the same plaintext "ex."

This demonstrates that Digraph Substitution Ciphers:

- Are still vulnerable to Frequency Analysis.
- Require frequency analysis to be conducted on pairs of letters instead of individual letters.

Security of Playfair Cipher

- Because it is done on pairs of letters, this Frequency Analysis is significantly harder to crack. Firstly, for a monoalphabetic cipher we have 26 possible letters to check.
- For a general Digraph Cipher we have 26 x 26 = 676 possible pairings we need to check in our frequency analysis. In the instance of the Playfair Cipher, we cannot encrypt to a double letter, so we remove the 26 possibilities of double letters, giving us 650 possible digraphs we need to check.
- By hand this task in monumental, but with the help of a computer, it can be done in a matter of seconds.

Security of Playfair Cipher

- Another useful weakness of the Playfair Cipher that can be exploited in cryptanalysis is the fact that the same pair of letters reversed will produce the same pair of letters reversed.
- For example, if the plaintext "er" encrypts to "HY", then the plaintext "re" will encrypt rto "YH". This is useful in some words in english such as "**de**part**ed**" which start and end with the same pair of letter in reverse.